

Project #3: Digital-Twin-Driven VLA Mobile Manipulator for Autonomous Strawberry Harvesting

NVIDIA Isaac Sim • Vision–Language–Action Control • Dexterous/Soft Grasping • Sim-to-Real Validation

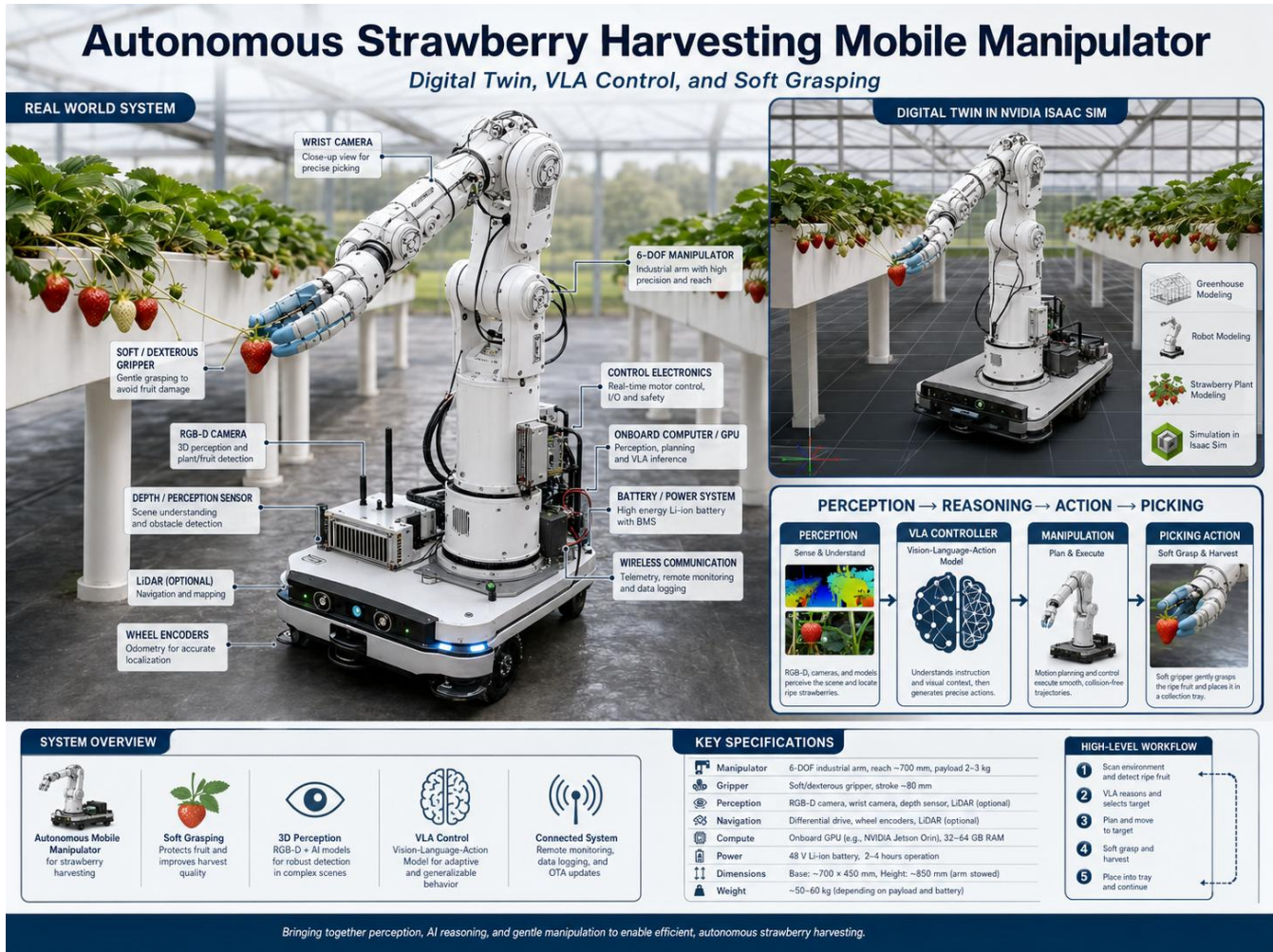


Figure 1. Proposed autonomous strawberry-harvesting mobile manipulator and its physical/digital-twin concept.

1. Project Description and International Collaboration

This graduation project will develop an autonomous mobile manipulator that can navigate inside a strawberry greenhouse, identify a high-quality ripe strawberry, approach it safely, harvest it using a compliant dexterous or soft gripper, and place it in a collection tray. The work continues and extends the existing Autonomous Strawberry Harvesting Robot platform by combining a physical robot, high-fidelity digital twins in NVIDIA Isaac Sim, multimodal perception, motion planning, and a Vision–Language–Action (VLA) controller. The primary research and implementation emphasis is the realistic modelling of strawberries and greenhouse scenes, the creation of reproducible digital-twin assets and scenarios, and the use of VLA-based reasoning to convert a task instruction and visual observations into validated robot actions before deployment on the developed robot.

The project will be conducted in collaboration with an agricultural-robotics laboratory at the **China National Engineering Research Center for Information Technology in Agriculture (NERCITA), China**. Regular online technical meetings will be arranged throughout the project with the Chinese team to review the greenhouse and strawberry models, simulation scenarios, perception and VLA methods, protocols, progress milestones, and final performance results. This collaboration is intended to expose the student team to international agricultural-robotics practice and strengthen the technical quality and relevance of the demonstrator.

2. Main Objectives

- Create reusable USD digital assets for the greenhouse, robot, cultivation tables, strawberry plants, fruit, stems and leaves.
- Model strawberry ripeness, geometry, occlusion, contact and grasp/detachment behaviour at an appropriate fidelity.
- Implement perception for fruit detection, ripeness assessment, 3D localization and target accessibility.
- Develop a VLA-guided task layer for target selection, action sequencing and recovery, linked to safe motion planning.
- Integrate the mobile base, 6-DOF arm, soft/dexterous gripper, cameras, navigation sensors and onboard computing.
- Validate the complete system in Isaac Sim and then on the physical robot using measurable sim-to-real metrics.

System Layout, Components and Control Architecture

Physical System Layout and Control Architecture

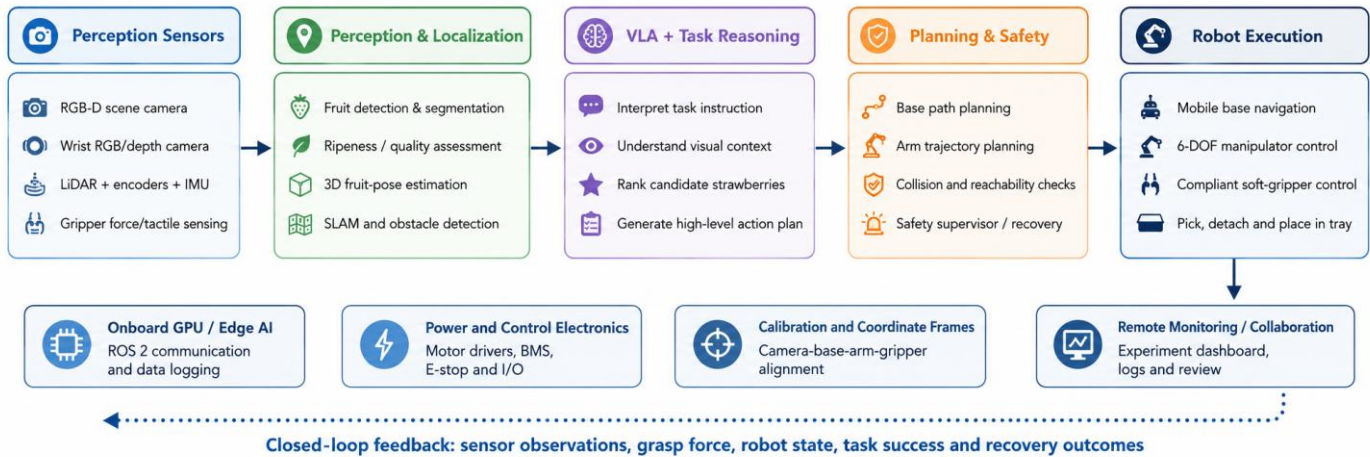
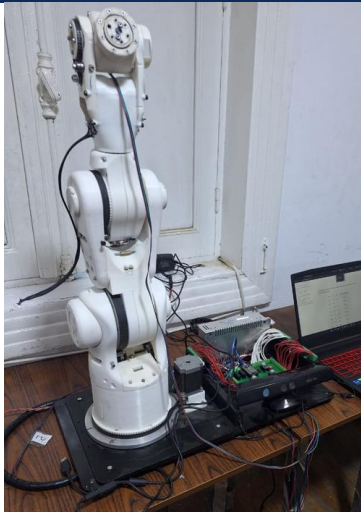


Figure 2. Big-picture system layout showing sensing, perception, VLA reasoning, planning, control and closed-loop feedback.

3. Physical Robot and Sensor Configuration



Existing 6-DOF arm and control-electronics prototype to be extended and integrated.

Subsystem	Planned Function
Mobile platform	Compact drive base, wheel encoders, navigation controller, battery, BMS, bumper/E-stop and provision for a collection tray.
Manipulator	Existing 6-DOF robotic arm; updated kinematic/dynamic model, calibrated joint limits and coordinated base-arm control.
End effector	Dexterous/soft gripper with compliant fingers and force/pressure/position feedback; optional stem-detachment or cutting feature.
Perception	Front RGB-D camera for scene mapping and fruit localization; wrist RGB/depth camera for close-range pose refinement and visual servoing.
Navigation and safety	LiDAR or depth-based obstacle sensing, wheel encoders, IMU, workspace limits, collision monitoring, safe stop and recovery behaviours.
Compute and software	Onboard GPU/edge computer, ROS 2 communication, perception and VLA inference, MoveIt-compatible planning, data logging and remote monitoring.

4. Closed-Loop Harvesting Operation

During operation, the mobile base localizes itself and moves to a harvesting zone. RGB-D and wrist-camera data are used to segment strawberries, estimate ripeness, determine the 3D pose of each fruit and assess whether a collision-free approach is possible. The VLA layer interprets the instruction (for example, “pick a ripe, undamaged and accessible strawberry”), combines it with the visual scene, ranks candidate fruits and proposes a high-level sequence. A deterministic planning and safety layer then verifies reachability, generates base and arm trajectories, limits interaction force and commands the gripper. Continuous camera, joint, navigation and gripper feedback is used to confirm grasping, detachment and placement, or to trigger a safe retry/recovery action.

Autonomous Strawberry Picking Sequence

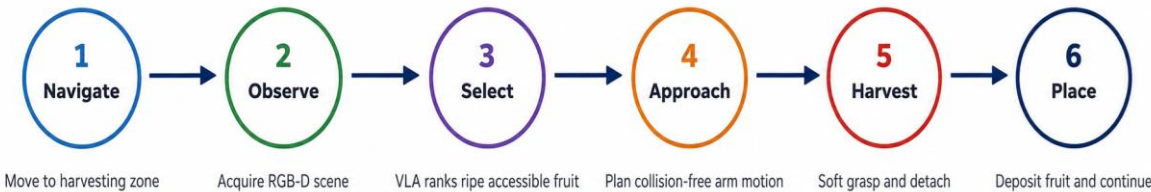


Figure 3. Operational sequence from greenhouse navigation to fruit placement and continuation.

Digital Twin, Strawberry Modelling and VLA Validation

NVIDIA Isaac Sim Digital Twin and VLA Development Pipeline

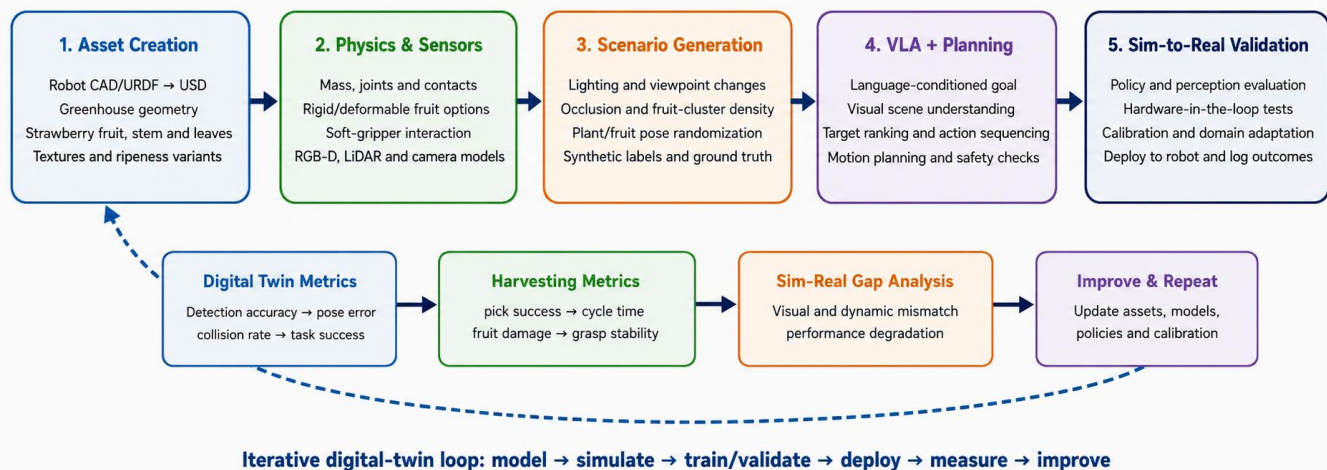


Figure 4. Digital-twin pipeline for asset creation, scenario generation, VLA validation and sim-to-real improvement.

5. NVIDIA Isaac Sim Digital-Twin Development

The digital twin will be the main development and verification environment before extensive physical trials. The robot CAD and kinematic model will be converted to a clean USD articulation with correct joint axes, limits, inertial properties, collision meshes and controller interfaces. The greenhouse model will include cultivation tables, crop rows, floor geometry, obstacles and realistic illumination. Sensor twins will reproduce RGB, depth, camera intrinsics, LiDAR/odometry and robot-state data so that the same ROS 2 interfaces can be used in simulation and on the real platform.

Priority Work Package: Strawberry and Plant Digital Assets

- Create parametrised fruit models covering size, shape, colour and multiple ripeness levels, together with stems, calyxes, leaves and realistic textures.
- Build fruit clusters with controllable pose, density, occlusion, accessibility and stem orientation; support procedural variation and real-to-simulation reconstruction from captured greenhouse data.
- Assign appropriate physical properties for contact, friction and grasping. Use simplified rigid models for large-scale training and higher-fidelity deformable/contact models for selected grasp and damage studies.
- Generate pixel-level masks, bounding boxes, depth, surface normals, 3D poses and ripeness labels automatically for perception training and evaluation.

6. VLA Control, Sim-to-Real Transfer and Validation

The VLA component will provide high-level embodied reasoning rather than bypassing conventional control and safety. It will fuse the task instruction with RGB/depth observations and structured robot state to identify the intended fruit, explain or encode the target choice, and issue constrained actions such as navigate, observe, re-position, approach, grasp, detach, place and recover. Each action will be checked by the motion planner, collision monitor and low-level controller. Isaac Sim will be used to evaluate the VLA system across large variations in lighting, fruit appearance, occlusion, camera viewpoint, base pose and plant layout. Domain randomization, synthetic data and recorded real images will reduce overfitting and support sim-to-real transfer.

Deployment will proceed through model-in-the-loop, software-in-the-loop and hardware-in-the-loop stages, followed by controlled physical picking trials. Camera calibration, coordinate-frame alignment, robot-system identification and real-image fine-tuning will be performed before autonomous operation. Real experiments will be logged and replayed in the twin to identify visual or dynamic mismatches and update the assets, policies and calibration parameters.

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